The Midnight Express Mystery

Whodunnit? · Angles & Parallel Lines · Teacher Guide and Answer Key

Unit	Time	Supplies	Group size
Angles & Parallel Lines	20–40 minutes	Pencil, scrap paper, scissors	Teams of 2–3 (or solo)

Skills practiced: Angle pairs with parallel lines, triangle angle sums, exterior angles and polygon angle sums.

General procedures

1. Give each student the **student sheet** (story and case board) and the **Show Your Work** page.
2. Print the **Clue Cards** (pages 3–7 of the student packet) and cut them apart on the dashed lines. Laminating them lets you reuse the set. Post them around the room.
3. Students visit the cards in any order. At each card they solve the problem on their work page, find their answer among the four choices, read the clue beside it, and cross things off the case board. **Every wrong choice leads to a false clue** that clears the real culprit, so a mistake shows up when the board does not narrow down to one suspect, one place and one method.
4. When a team thinks it knows, they bring you the three names. If they are wrong, let them hunt for the card where something went wrong.

Solution: Dr. Marsh took it, in the Caboose, using this method: Locked in a violin case.

The reveal (read aloud): When the train entered the tunnel, **Dr. Marsh** slipped into the **Caboose**, locked the Blue Diamond in a **violin case**, and carried it off the train at the next station. Every alibi and method you cleared led to the doctor. All aboard for justice!

Clue map: what each correct clue clears

Card	Clears
1	Suspect: Conductor Hale · Place: Engine Car
2	Method: Hidden in a suitcase
3	Suspect: Porter Bell · Place: Baggage Car
4	Method: Sewn into a hat
5	Suspect: Madame Vesper · Place: Sleeping Car
6	Method: Dropped from the window
7	Suspect: Mr. Quint · Place: Observation Deck
8	Method: Folded in a newspaper
9	Suspect: Ms. Iris · Place: Dining Car
10	Method: Buried in the coal bin

Answer Key

The Midnight Express Mystery · Correct choice and worked solution for each card

#	Problem	Correct choice	Worked solution	Clue it leads to
1	Parallel lines are cut by a transversal. Alternate interior angles measure (3x + 10)° and (5x − 20)°. Find the measure of each angle.	**D.** 55°	3x + 10 = 5x − 20, so x = 15 3(15) + 10 55°	Conductor Hale was driving the Engine Car. The engineer saw him the whole time.
2	Parallel lines are cut by a transversal. Same-side interior angles measure (2x + 20)° and (3x − 10)°. Find the larger angle.	**D.** 92°	5x + 10 = 180, so x = 34 angles: 88° and 92° 92°	Every suitcase was searched at the station. None held the Diamond.
3	Parallel lines are cut by a transversal. Corresponding angles measure (4x − 15)° and (2x + 25)°. Find the measure of each angle.	**A.** 65°	4x − 15 = 2x + 25, so x = 20 4(20) − 15 65°	Porter Bell was sorting luggage in the Baggage Car with a helper beside him.
4	Parallel lines are cut by a transversal. Alternate exterior angles measure (6x − 4)° and (4x + 16)°. Find the measure of each angle.	**B.** 56°	6x − 4 = 4x + 16, so x = 10 6(10) − 4 56°	The fancy hat was cut open and flattened. No diamond sewn inside.
5	The angles of a triangle measure 2x°, (3x + 5)° and (4x − 14)°. Find the largest angle.	**A.** 70°	9x − 9 = 180, so x = 21 angles: 42°, 68°, 70° 70°	Madame Vesper was resting in the Sleeping Car. Her maid never left her side.
6	An exterior angle of a triangle measures (5x + 3)°. The two remote interior angles measure (2x + 30)° and (x + 21)°. Find the exterior angle.	**B.** 123°	5x + 3 = (2x + 30) + (x + 21) 2x = 48, so x = 24 5(24) + 3 123°	The windows were all latched shut during the trip, so nothing was dropped out.
7	Two vertical angles measure (7x − 9)° and (5x + 17)°. Find the measure of each angle.	**C.** 82°	7x − 9 = 5x + 17, so x = 13 7(13) − 9 82°	Mr. Quint was admiring the stars from the Observation Deck with several passengers.
8	Angles measuring (x + 40)° and (2x − 10)° are supplementary. Find the measure of the second angle.	**D.** 90°	3x + 30 = 180, so x = 50 2(50) − 10 90°	The newspaper on the seat is a plain one with nothing folded inside.
9	The vertex angle of an isosceles triangle measures 44°. Find the measure of each base angle.	**C.** 68°	base angles are equal: 2b + 44 = 180 2b = 136 68°	Ms. Iris was serving dessert in the Dining Car. The passengers all remember it.
10	Find the sum of the interior angles of a hexagon.	**A.** 720°	sum = (n − 2) · 180 with n = 6 4 · 180 720°	The coal bin was shoveled out. Only coal and soot.

Wrong-Answer Map

The Midnight Express Mystery · What a wrong choice looks like, and the false clue it leads to

Card	Wrong choice	Likely error	False clue
1	A. 125°	Treated alternate interior angles as supplementary	The violin case was opened at the station. Just a violin and a spare string.
1	B. 15°	Gave x instead of the angle	Dr. Marsh was checking the engineer’s pulse in the Engine Car. The engineer says he stayed the whole hour.
1	C. 110°	Added the two angles	Conductor Hale was waving his lantern from the Caboose. The brakeman waved back.
2	A. 34°	Gave x instead of the angle	Porter Bell was sweeping out the Caboose. The pile of dust is still by the door.
2	B. 80°	Set the angles equal instead of supplementary	The only violin case on board is dented and stuck shut. Nothing could be locked inside.
2	C. 88°	Gave the smaller angle	Dr. Marsh was sipping tea in the Dining Car. The waiter refilled his cup twice.
3	B. 115°	Treated corresponding angles as supplementary	The violin case is lined with velvet and holds only sheet music.
3	C. 130°	Added the two angles	Madame Vesper was practicing scales in the Caboose. The brakeman covered his ears.
3	D. 20°	Gave x instead of the angle	Dr. Marsh was tucking a sleepy child in at the Sleeping Car. The mother thanked him.
4	A. 112°	Added the two angles	Mr. Quint was polishing rings in the Caboose. A customer bought one right there.
4	C. 124°	Treated alternate exterior angles as supplementary	The latch on the violin case is broken. It could not have been locked.
4	D. 10°	Gave x instead of the angle	Dr. Marsh was lifting trunks in the Baggage Car. Porter Bell kept count of every one.
5	B. 21°	Gave x instead of the angle	Ms. Iris was hanging up her apron in the Caboose. The hook still holds it.
5	C. 42°	Gave the smallest angle	The violin case was weighed at the station. It was exactly as heavy as the violin alone.
5	D. 68°	Gave the middle angle	Dr. Marsh was pointing out constellations on the Observation Deck. Three kids took notes.
6	A. 24°	Gave x instead of the exterior angle	Dr. Marsh was handing the engineer a cough drop in the Engine Car just before the whistle blew.
6	C. 78°	Gave one remote interior angle	The violin case rode in the overhead rack the whole trip. Its lock was never turned.
6	D. 57°	Subtracted from 180 (used the interior angle)	Conductor Hale was punching tickets in the Caboose. The stubs are stamped with the time.
7	A. 164°	Added the two vertical angles	Porter Bell was napping on a bench in the Caboose. His snoring was heard down the track.
7	B. 98°	Treated vertical angles as supplementary	The violin case key was found in the lost-and-found box. No one used it that night.
7	D. 13°	Gave x instead of the angle	Dr. Marsh was carving roast beef in the Dining Car. Ms. Iris handed him the knife.
8	A. 130°	Gave the supplement of x	The violin case smells only of rosin. Nothing else has ever been inside it.

8	B. 30°	Used complementary (90) instead of supplementary	Madame Vesper was gazing at the tracks from the Caboose. Her scarf blew off the back.
8	C. 50°	Gave x instead of the angle	Dr. Marsh was reading by lamplight in the Sleeping Car. His bunkmate saw him until dawn.
9	A. 136°	Subtracted from 180 but did not halve	Dr. Marsh was mending a torn suitcase in the Baggage Car. The needle and thread are still there.
9	B. 88°	Doubled the vertex angle	The violin case was tapped and listened to. It sounded completely hollow.
9	D. 46°	Used 90 instead of 180	Mr. Quint was sorting his business cards in the Caboose. The brakeman took one.
10	B. 1080°	Used 8 triangles (octagon)	Dr. Marsh was watching the sunrise from the Observation Deck with a group of passengers.
10	C. 360°	Used the exterior angle sum	The violin case was dusted for fingerprints. Only the owner’s were found.
10	D. 540°	Used $(n-3)\cdot 180$ (pentagon)	Ms. Iris was stacking dessert plates in the Caboose. Not one plate was chipped.